

ChE 205 – Computational Methods in Chemical Engineering – Fall 2018

Student Learning Outcome 1

Rubric for assessment problem on final exam (Problem # 1c, 3, 6, and 2a)

Application of computational methods in solving chemical engineering problems.

SLO 1 Performance Indicator (PI)	(1) Unsatisfactory	(2) Developing	(3) Satisfactory
1. Apply mathematical and scientific principles to formulate models and systems relevant to the subject Problem 1c, 8 pts	Able to successfully combines mathematical and/or scientific principles to formulate models and systems relevant to chemical engineering (<4 pts)	Chooses a mathematical model or scientific principle that applies to an engineering problem, but has trouble in model development (4-6 pts)	Does not understand the connection between mathematical models and the system or process to be analyzed or designed (>6 pts)
2. Solve engineering problems by using the concepts of algebra, calculus, differential equations and/or linear algebra Problem 3, 10 pts	applies concepts of algebra, calculus, differential equations and/or linear algebra to solve chemical engineering problems (<4 pts)	Shows nearly complete understanding of applications of calculus and/or linear algebra in problem-solving (4-7 pts)	Does not understand the application of algebra, calculus, differential equations and/or linear algebra in solving chemical engineering problems (>7 pts)
4. Translates academic theory into engineering applications Problem 6, 5 pts	Translates academic theory into engineering applications and accepts limitations of mathematical models of physical reality (< 2 pts)	Some gaps in understanding the application of theory to the problem and expects theory to predict reality (2-3 pts)	Does not appear to grasp the connection between theory and the problem (>3 pts)
6. The uses of appropriate resources needed to solve problems Problem 2a, 4 pts	Uses no resources to solve problems (< 2 pts)	Uses limited resources to solve problems (2-3 pts)	Uses appropriate resources to locate information needed to solve problems (>3 pts)

Averages:

Performance Indicators: (1) 2.48 (2) 1.66 (4) 2.52 (6) 2.55

Overall Average: 2.30

Standard Deviation: 0.37

Average scores were fairly uniform across Performance Indicators (PI) 1, 4, and 6. A rather low score was observed in PI 2, designated for problem 3. Also, there was wide variation of PI among students. This data suggests that most students were strong in (i) applying mathematical and scientific principles to formulate models, (ii) translate

academic theory into engineering applications, and (iii) uses of appropriate resources needed to solve problems; however they were weak in solving engineering problems using differential equations and linear algebra (16/29 were unsatisfactory to solve problem 2).

Problem 1 (10 pts). Consider the following algebraic equation:

$$x = -2 \log_{10}(ax + b), \quad (1)$$

where $a = 0.0006$, and $b = 0.0135$.

- To find the root of above equation, you need to convert that to a function. Plot the function whose root you are trying to find. (1 pts)
- Based on your results in part a, make two good initial guesses, x_1 and x_2 . (1 pt)
- Write a VBA program to solve for x , using secant method, and the initial guesses, x_1 and x_2 , you made in part b. The VBA script should take a , b , x_1 , x_2 , and the tolerance as inputs through the spreadsheet. Then it should output
 - the solution, x ,
 - the total number of iterations, and
 - the error at each iteration,

as shown below. (8 pts)

	A	B	C	D	E	F
1	0.0006	=a			iteration	error
2	0.0135	=b			1	????
3	????	=x1			2	????
4	????	=x2			3	????
5	1.00E-09	=tol			4	????
6	????	=x			5	????
7	????	=iterations			6	????
8					7	????
9					8	????

Figure 1: Sample input and output of VBA script in Problem 1.

Problem 2 (12 pts). The "Monod" model for bioreaction kinetics can be expressed as:

$$\begin{aligned}\frac{ds}{dt} &= -\frac{k_1sx}{k_2+s}, \\ \frac{dx}{dt} &= a\frac{k_1sx}{k_2+s} - bx,\end{aligned}$$

where

s = growth limiting substrate concentration (ML^{-3})

x = biomass concentration (ML^{-3})

k_1 = maximum specific uptake rate of the substrate (T^{-1}) = 5

k_2 = half saturation constant for growth (ML^{-3}) = 20

a = yield coefficient (MM^{-1}) = 0.05

b = decay coefficient (T^{-1}) = 0.01

Initial conditions are $s(0) = 1000$, and $x(0) = 100$, and use $\Delta t = 0.005$.

- (a) Solve this ODE-IVP using forward Euler in spreadsheet (this is not VBA programming). Plot s and x in red and blue, respectively, vs $0 \leq t \leq 2$. (4 pts)

Problem 3 (10 pts). Using the Crank-Nicholson method, write a VBA program to solve the following heat equation with a heat sink (the second term on the right hand side):

$$\frac{\partial T}{\partial t} = \frac{\partial^2 T}{\partial x^2} - T, \quad (2)$$

subject to the following initial and boundary conditions

$$T(x, t = 0) = 0, \quad \frac{\partial T(x = 0, t)}{\partial x} = -1, \quad T(x = 0.5, t) = 0, \quad (3)$$

where T is the temperature, $0 \leq t \leq 0.1$ is the time variable, and x is the spatial coordinate. Use $\Delta x = 0.025$, and $\Delta t = 0.002$. Plot $T(x, t = 0.1)$ against $0 \leq x \leq 0.5$.

Problem 6 (5 pts). Consider the following regression

$$\ln(y) = -c_1 \exp\left(-\frac{c_2}{T}\right)t, \quad (6)$$

where $\ln$ is the natural *log*, y is the dependent variable, t and T are the two independent variables. c_1 and c_2 are two parameters to be determined. A typical set of data appears below.

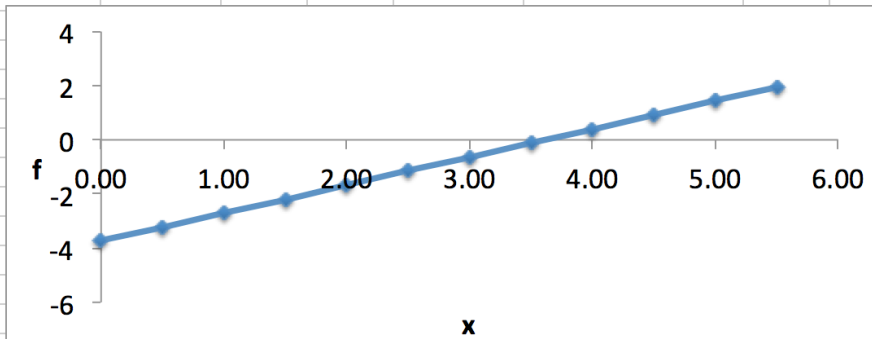
t	T	y
0.10	100	0.980
0.20	100	0.983
0.10	200	0.544
0.50	200	0.225
0.02	300	0.566
0.06	300	0.034

Find c_1 and c_2 by minimizing the sum of residual squared between y^{data} and y^{calc} . (5 pts)

Solutions

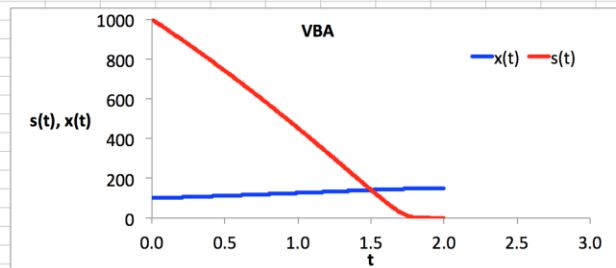
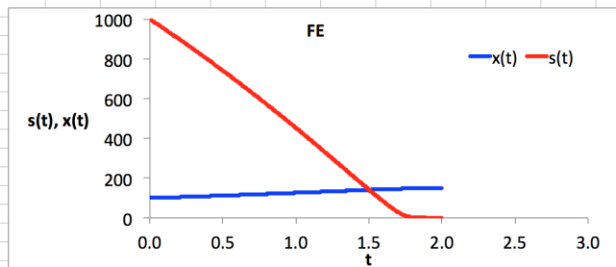
Problem 1c:

A	B	C	D	E	F	G	H	I	J
0.0006	=a			iteration	error			x	f
0.0135	=b			1	-6.09E-01			0.00	-3.74
2	=x1			2	-6.41E-04			0.50	-3.22
3	=x2			3	-2.45E-07			1.00	-2.7
1.00E-09	=tol			4	-9.67E-14			1.50	-2.18
3.6101	=x							2.00	-1.67
4	=iterations							2.50	-1.15
								3.00	-0.63
								3.50	-0.11
								4.00	0.403
								4.50	0.919
								5.00	1.435
								5.50	1.951

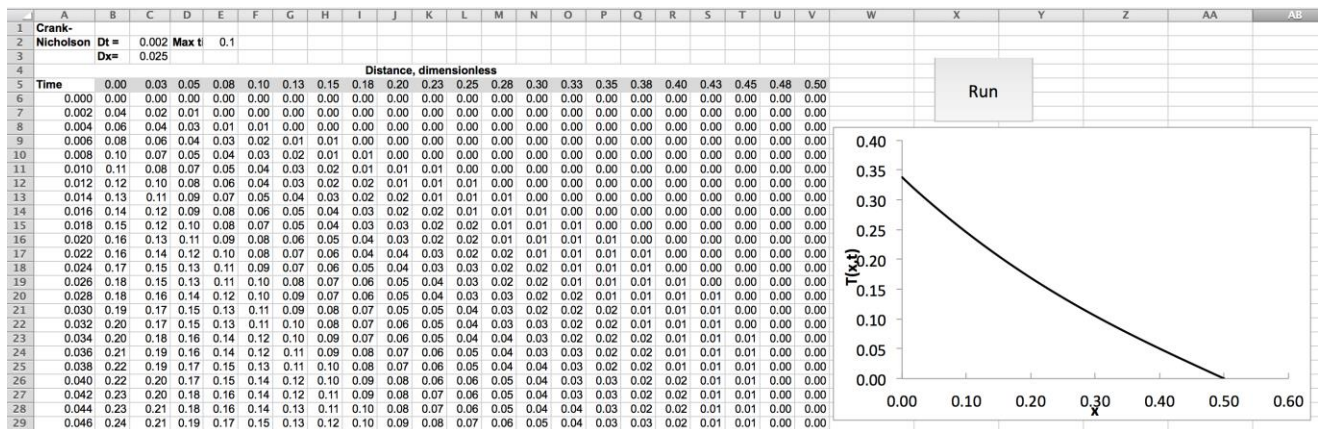


Problem 2a:

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
1	tmax=	2																
2	dt=	0.005																
3	r1=	5																
4	r2=	20																
5	a=	0.05																
6	b=	0.01																
7	t	s	x															
8	0.000	1000.00	100.00															
9	0.005	997.55	100.12															
10	0.010	995.10	100.24															
11	0.015	992.64	100.35															
12	0.020	990.18	100.47															
13	0.025	987.72	100.59															
14	0.030	985.25	100.71															
15	0.035	982.79	100.83															
16	0.040	980.31	100.94															
17	0.045	977.84	101.06															
18	0.050	975.37	101.18															
19	0.055	972.89	101.30															
20	0.060	970.41	101.42															
21	0.065	967.92	101.54															
22	0.070	965.43	101.66															
23	0.075	962.94	101.78															
24	0.080	960.45	101.90															
25	0.085	957.96	102.02															
26	0.090	955.46	102.14															
27	0.095	952.96	102.26															
28	0.100	950.45	102.38															
29	0.105	947.95	102.50															
30	0.110	945.44	102.62															



Problem 3



Problem 6

	A	B	C	D	E	F	G
1	c1=	2135.94	c2=	1235.47			
2		t	T	y	ycalc	residual^2	
3		0.1	100	0.980	0.999	0.000	
4		0.2	100	0.983	0.998	0.000	
5		0.1	200	0.544	0.642	0.010	
6		0.5	200	0.225	0.109	0.013	
7		0.02	300	0.566	0.499	0.004	
8		0.06	300	0.034	0.124	0.008	
9	sum					0.036	
10							